

Samuel Fernandes, PhD

Senior Technical Program Manager | Advanced Energy Technologies & Technical Diligence

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Professional Summary

Technical Program Manager and energy technologist with **15+ years** of experience leading R&D, technology assessment, and commercialization of advanced energy technologies. Direct hands-on experience taking a national-lab innovation from early-stage prototype through First-of-a-Kind (FOAK) deployment and licensed commercial product (Sensor Suitcase, FLC Award for Excellence in Technology Transfer). Deep expertise in Technology Readiness Level (TRL) evaluation, technical due diligence, performance-claim validation, manufacturing scalability, grid integration of distributed and clean energy resources, and translating engineering tradeoffs into executive-level investment and partnership decisions. Trusted partner to utilities, national labs, DOE programs, startups, and academic institutions across the global clean energy ecosystem.

Core Competencies

Technology Assessment & Diligence: TRL evaluation | FOAK deployment risk | Performance-claim validation | Manufacturing & cost-down analysis | Independent Engineer (IE) coordination | Project execution risk
Roadmapping & Strategy: Multi-sector technology roadmaps (advanced nuclear, next-gen geothermal, long-duration storage, demand flexibility) | Cross-functional capability planning | Procurement & investment strategy support
Power Systems & Grid Integration: Grid emissions modeling | Hourly/marginal/locational analysis | DER and demand-side integration | Scope 2 carbon accounting | eGrid, WattTime, Cambium, AVERT
Program Leadership: Cross-functional execution | Executive communication | DOE & utility partnerships | Commercial agreement support | Technical writing & reporting

Professional Experience

Energy Technology Researcher / Technical Program Manager

Lawrence Berkeley National Laboratory

January 2019 – Present

Building Technology and Urban Systems Division

Technology Assessment, TRL Evaluation & Technical Due Diligence:

- Performed technical assessments of emerging energy and grid-edge technologies, evaluating TRL, performance claims, manufacturing scalability, and pathways to cost reduction across building-grid, DER, and demand-flexibility solutions
- Conducted project-level technical diligence on engineering designs, technical milestones, project delivery plans, and execution risks for DOE-funded and utility-funded deployments, surfacing risks before capital commitment
- Built quantitative diligence frameworks combining hourly performance data, regional grid signals (eGrid, WattTime, Singularity, AVERT, Cambium through 2050), and scenario analysis to stress-test technology and project performance under future-grid conditions
- Translated complex engineering and grid-integration tradeoffs into decision-ready insights for funders, program owners, and executive stakeholders evaluating clean-energy investments

Technology Roadmapping & Strategy:

- Developed multi-year technology roadmaps for building-grid integration, demand flexibility, and decarbonization pathways, aligning technical capabilities with commercial deployment, sustainability, and procurement objectives
- Defined long-horizon scenario analyses (through 2050) showing how the value and viability of clean-energy actions evolve as regional grids decarbonize, informing long-duration investment and procurement strategy
- Evaluated tradeoffs across emissions accounting methods (annual average, hourly, marginal, non-baseload) to define defensible metrics for carbon-free energy strategy

Cross-Functional Program Leadership:

- Led cross-functional, multi-organization programs spanning utilities, national labs, DOE program offices, and academic partners; planned requirements, identified technical risks, managed schedules, and communicated clearly with all stakeholders
- Managed DOE- and partner-funded programs end-to-end—scoping, technical execution, publication, and stakeholder communication—with contributors across multiple disciplines and institutions
- Authored peer-reviewed papers, technical reports, and executive briefings translating model behavior and technology assessments into concise, decision-ready recommendations

Technical Program Manager – Energy Technologies

Lawrence Berkeley National Laboratory

October 2014 – December 2018

Energy Technologies Area

First-of-a-Kind (FOAK) Technology Development & Commercialization – Sensor Suitcase:

- Core team member taking the **Sensor Suitcase**—a portable retro-commissioning hardware/software platform invented at LBNL—from early prototype (TRL 3–4) through field validation, FOAK deployment, and licensed commercial product (TRL 8–9)

- Led technical milestones, field-validation campaigns, and deployment risk reduction across multiple U.S. climate zones and building stock; quantified performance claims, manufacturability, and cost-down trajectory to support transition to industry
- Coordinated technology transfer through the DOE Lab Technology to Market (Lab-to-Market) program, structuring the technical foundation of the licensing agreement and ensuring performance guarantees were technically sound—work recognized by the **2020 Federal Laboratory Consortium Award for Excellence in Technology Transfer**

Utility Partnerships, Grid Integration & Diligence:

- Managed technical programs with major U.S. utilities (Seattle City Light, Bonneville Power Administration, Eversource Energy) on EMIS, building-grid integration, and deployment strategy, defining technical requirements for integrating advanced energy and demand-side technologies into utility operations
- Led DOE- and utility-funded R&D on commercial-building analytics, monitoring-based commissioning, and emerging energy-efficiency technology evaluation; provided the technical foundation for partnership and procurement decisions
- Secured multi-million-dollar program funding through proposal leadership and clear articulation of technical, commercial, and grid-integration value to utility and federal stakeholders

Manager of Sustainability and Green ICT Strategy

July 2011 – September 2014

Australian National University

- Oversaw building and ICT data management for a 130-building research campus, including technical assessment and PUE quantification for two large data centers—directly relevant to hyperscale energy use, efficiency, and carbon-free energy strategy
- Led campus-wide ICT carbon-footprint assessment, establishing emissions accounting and decarbonization roadmap for a complex multi-stakeholder organization

Co-Founder & CTO

May 2011 – August 2013

MicroEnergyLabs Pty. Ltd.

Acquired by Active Grid Pty. Ltd.

- Developed a novel optical sensing technology for real-time electricity consumption, advancing it from concept through prototype, pilot deployment, and acquisition—direct experience navigating the technical and commercial challenges of hardware commercialization
- Selected for the IgnitionLabs Clean Technology Accelerator; led product strategy, technical partnerships, and market validation through to successful exit

Education

PhD, Civil and Environmental Engineering	2025
<i>UC Berkeley</i>	<i>Minor: Computer Science and Machine Learning</i>
Master of Science, Civil and Environmental Engineering	2020
<i>UC Berkeley</i>	
Clean Energy Leadership Institute Fellow	2017
<i>First Bay Area cohort</i>	<i>Clean energy ecosystem expertise and energy justice principles</i>
Master of Engineering, Systems Engineering	2010
<i>Australian National University</i>	
Bachelor of Engineering, Electronics and Computer Science	2008
<i>University of Mumbai</i>	

Selected Technical & Programmatic Highlights

- FOAK Technology Transfer (Sensor Suitcase)** – Took a national-lab energy technology from prototype through licensed commercial product; recognized with the 2020 Federal Laboratory Consortium Award for Excellence in Technology Transfer
- Forward-Looking Grid Diligence** – Applied Cambium 2050 projections and multi-region grid datasets to stress-test the long-term viability of energy technologies and procurement strategies under decarbonizing grids
- Time- & Locational-Value Analytics** – Quantified how time-of-day, season, and region materially shift carbon and economic outcomes of clean-energy actions, informing technically sound performance guarantees
- Ecosystem Network** – Active relationships across DOE national labs, U.S. utilities (SCL, BPA, Eversource), academic institutions, and the clean-energy startup ecosystem (CELI alumni network)

Awards & Recognition

- 2020 Federal Laboratory Consortium Award for Excellence in Technology Transfer** – Sensor Suitcase Retro-commissioning Commercialization Team
- 2015 Davis Projects for Peace Prize** – Off-grid solar energy deployment, Mozambique
- 2011 Innovation ACT Prize** – Best business plan, MicroEnergyLabs

Selected Publications

“Using advanced M&V data to quantify the greenhouse gas impact of energy efficiency projects in commercial buildings” – 2025. Hourly emissions accounting and Scope 2 methodology for project-level GHG quantification

“Meter-Based Assessment of the Time and Locational Benefits of a Large Utility’s DSM Portfolio” – *ACEEE Summer Study on Energy Efficiency in Buildings*, 2020

“Accuracy of hourly energy predictions for demand flexibility applications” – *Energy and Buildings*, 2023

“Exploring the GHG implications of electric vehicles in California” – *Energy Policy*, 2019. Grid emissions and electricity-system impacts under evolving generation mix

“Gradient boosting machine for modeling the energy consumption of commercial buildings” – *Energy and Buildings*, 2018

16+ peer-reviewed publications spanning advanced energy technologies, grid emissions, decarbonization, and applied energy analytics

Technical Skills

Technology Diligence: TRL frameworks | FOAK deployment risk | Performance validation | Cost & manufacturability analysis | Project execution risk | IE coordination

Energy Systems Domain: Advanced nuclear (familiarity) | Next-gen geothermal (familiarity) | Long-duration energy storage | Demand flexibility | DER integration | Commercial buildings & data centers

Grid & Carbon Data: eGrid | WattTime | Singularity | AVERT | Cambium | Hourly & marginal emissions | Scope 2 accounting

Analytics: Python (pandas, numpy, scipy) | SQL | Time-series & scenario analysis | Reproducible workflows | Executive reporting

Global Impact & Volunteering

Climate Change AI Summer School Teaching Assistant | **2023** – AI for Energy Systems and AI for Buildings & Cities curriculum

Solar Energy System Designer & Installer | **Mozambique** | **2015** – Off-grid solar serving a school of 600 students